

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Sairaj C	Reg. No.:	192411018
Section / Batch:	BE-CSE	Date:	23/072026

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1

SECTION A – Technical Presentation

Code of Conduct

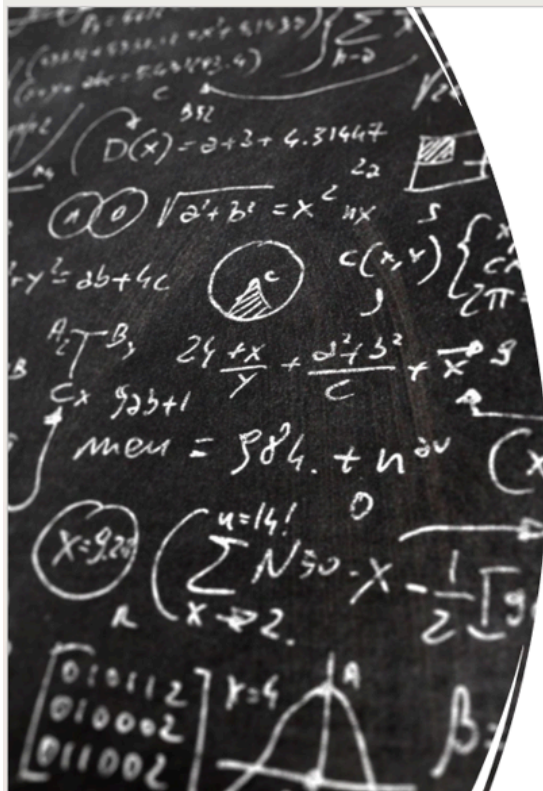
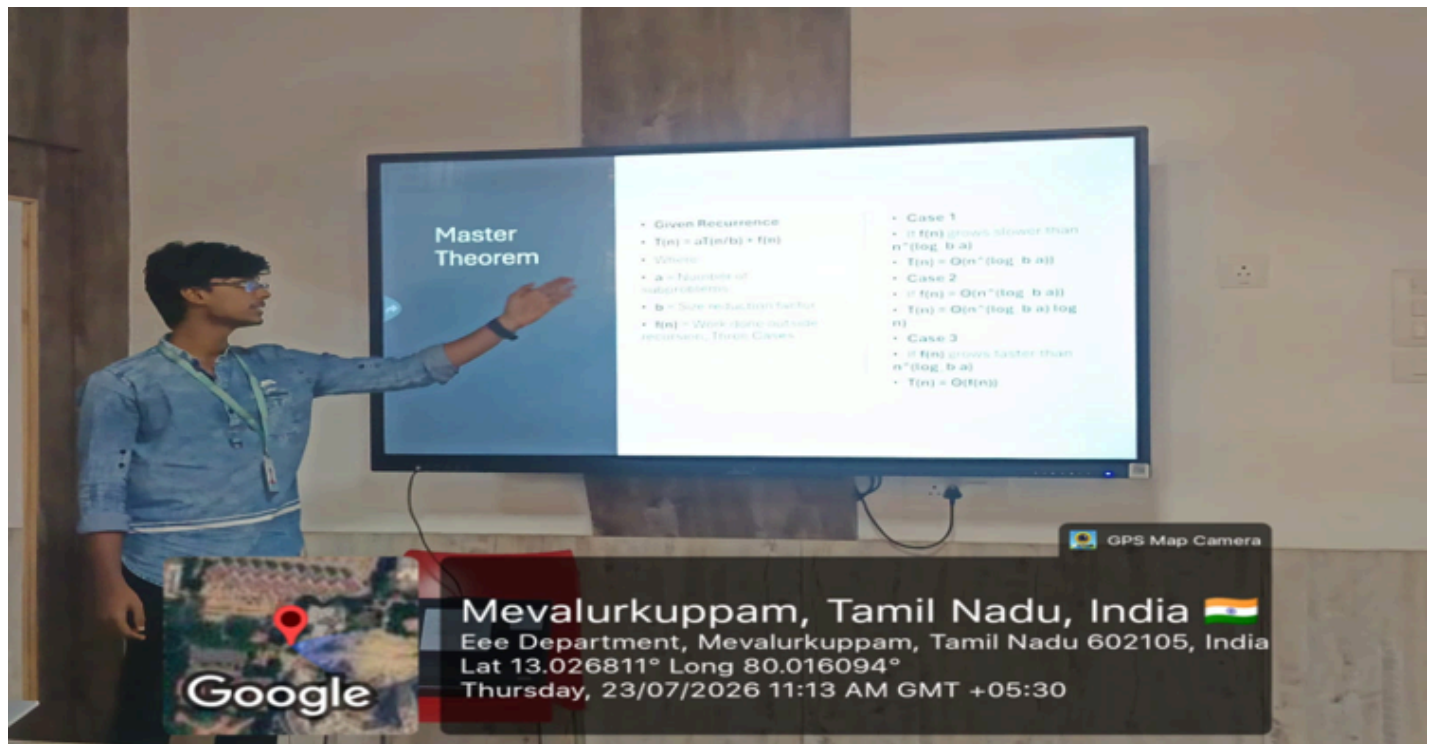
I Sairaj C(192411018) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sairaj C

Q1. Master Theorem Application

Consider the recurrence relation $T(n) = 2T(n/2) + n$. Analyze the problem using the Master Theorem and determine its time complexity. Clearly explain each case of the theorem and justify the selected case. Use tools such as recursion tree diagrams for better understanding. Present the solution clearly with proper

interpretation of results.



Master Theorem Application

Presented by:

SAIRAJ C

Reg. No.: 192411018

Master Theorem

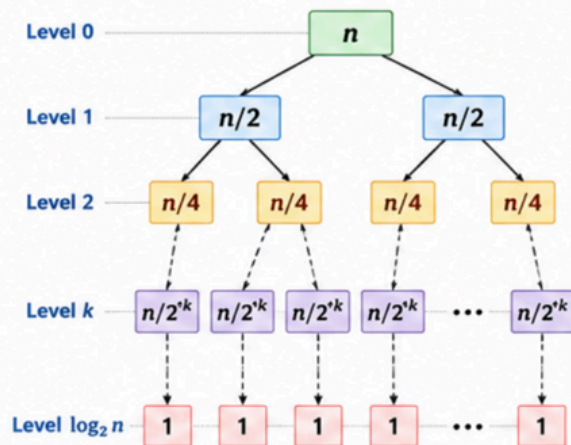
- **Given Recurrence**
- $T(n) = aT(n/b) + f(n)$
- Where:
 - a = Number of subproblems
 - b = Size reduction factor
 - $f(n)$ = Work done outside recursion, Three Cases
- **Case 1**
 - If $f(n)$ grows slower than $n^{\log_b a}$
 - $T(n) = \Theta(n^{\log_b a})$
- **Case 2**
 - If $f(n) = \Theta(n^{\log_b a})$
 - $T(n) = \Theta(n^{\log_b a} \log n)$
- **Case 3**
 - If $f(n)$ grows faster than $n^{\log_b a}$
 - $T(n) = \Theta(f(n))$

Applying Master Theorem

- **Recurrence**
- $T(n) = 2T(n/2) + n$
- Identify the values:
 - $a = 2$
 - $b = 2$
 - $f(n) = n$
- **Calculate:**
 - $n^{\log_2 2} = n^1 = n$
 - Since
 - $f(n) = \Theta(n)$
 - This matches Case 2.
 - Time Complexity
 - $T(n) = \Theta(n \log n)$

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Recursion Tree



Cost at each level

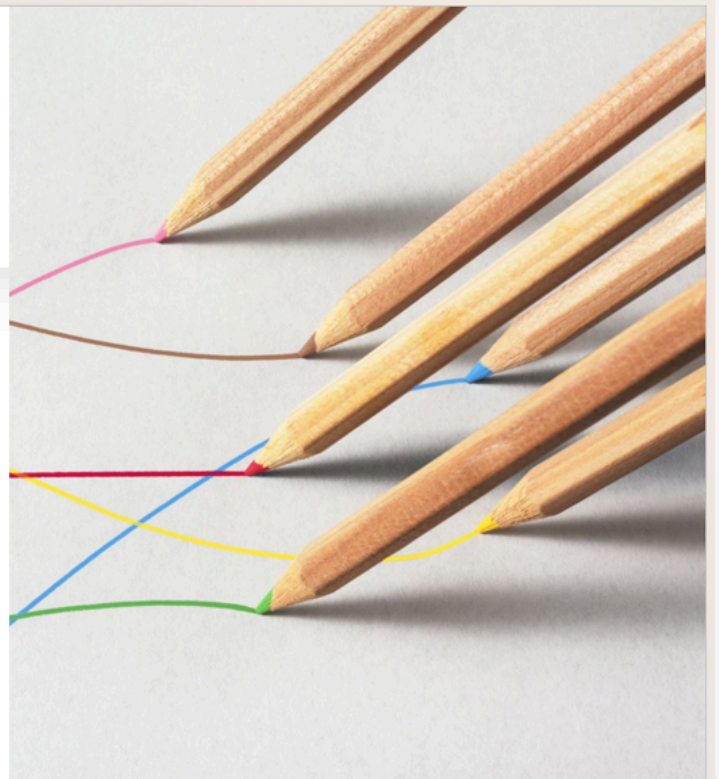
- Level 0 = n
- Level 1 = n
- Level 2 = n
- ...
- Total levels = $\log_2 n$

Total Cost
 $n \times \log n$

Overall Complexity
 $= \Theta(n \log n)$

Interpretation & Conclusion

- **Interpretation**
- The problem is divided into **2 equal subproblems**.
- Each recursive level performs **linear work (n)**.
- There are **$\log_2 n$** recursive levels.
- Therefore, the total work is **$n \log n$** .

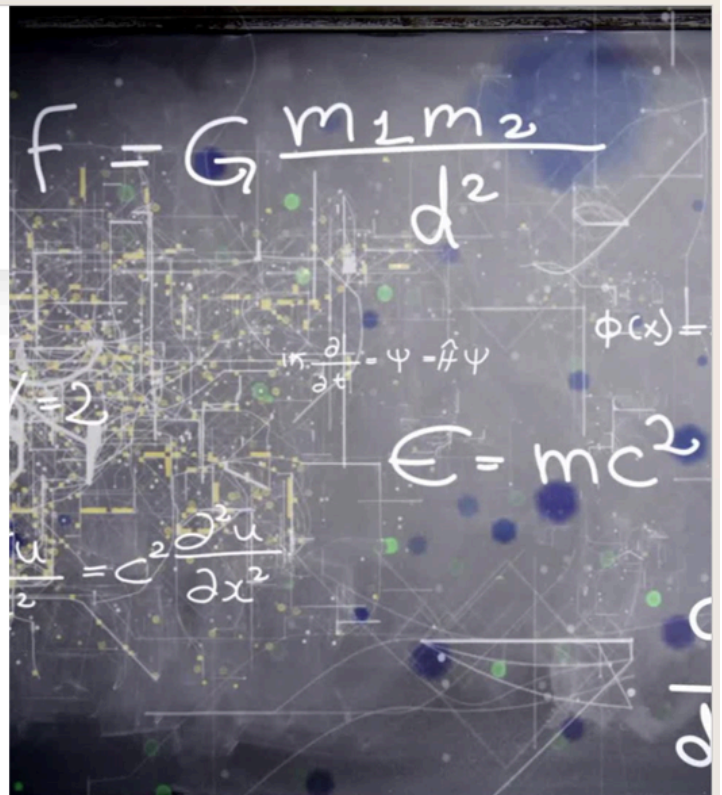


Conclusion

- The recurrence satisfies **Master Theorem Case 2**.
- The algorithm is efficient for large inputs.
- Many divide-and-conquer algorithms, such as **Merge Sort**, have this same timecomplexity.

Final Result

- Time Complexity = $\Theta(n \log n)$



RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%	Highly accurate, in depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25%	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20%	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communi cation	15%	Highly engaging, confident, and audience focused delivery	Clear, organized, and professional communication	Understandab le but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professio nalism	10%	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalis m; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30%				
Problem Identification	25%				
Use of Tools	20%				
Communication	15%				
Professionalism	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____